EXPERIMENT REPORT: FULL WAVE RECTIFIER CIRCUIT WITHOUT AND WITH FILTER

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Abstract

The full-wave bridge rectifier was constructed on a breadboard using four identical diodes, wires and an optional capacitor for filter. The input AC voltage is taken from a step-down transformer connected to the power supply, giving an output voltage of 8.00 volts. The output DC voltage V_{dc}, output AC voltage (RMS) V_{ac} and the input AC current (RMS) I_{in} are measured for four different load resistances (0.549Ω, 1.008Ω, 4.68Ω and 9.88Ω) for without filter and with filter ($C = 9.68\mu$F, $C = 16.72\mu$F and $C = 39.5\mu$F). It is found that the full-wave bridge rectifier can rectify the given AC supply into a DC supply across the load resistance. The ripple factor decreases with an increase in the capacitance (filter) for a constant resistance. However, it increases with the resistance for without filter, but decreases for with filter-case.

1 Objectives

- To construct a full-wave bridge rectifier and analyse its output.
- To see the affect of a capacitor in shunt with the load on the output signal.

2 Theory and Working Formula

A rectifier is a circuit that converts an AC voltage into direct current. Rectifiers can be a half-wave or a full-wave rectifier. Half-wave rectifiers can rectify only half of the AC supply whereas the other half is completely blocked by it. For a completely rectified signal, we need to use a full-wave rectifier. The full-wave rectifier can be of two types: one where two diodes and centre-tapping is required, and the other which is also known as a bridge rectifier uses 4 diodes. The rectified signals from the half-wave and full-wave rectifier are given in figure 2 as predicted by theory.

The circuit diagram for the full-wave bridge rectifier is given in figure 1. In a full-wave bridge rectifier, four diodes are used with two of them functioning in each half-cycle of the AC current. In one of the half-cycles, diodes D_1 and D_2 are in forward bias and D_3 and D_4 in reverse bias, whereas in the other half-cycle, diodes D_3 and D_4 are in forward bias and the other two in reverse bias.

For a full-wave rectifier, the average DC output is given by:

$$V_{\text{dc}} = \frac{2V_{\text{max}}}{\pi} \tag{1}$$

where V_{dc} is the output DC voltage and V_{max} is the amplitude of the AC voltage.

The ripple factor of the rectified signal is a measure of the stability of the output DC signal. It is defined as:

$$r = \frac{V_{\text{ac}}}{V_{\text{dc}}} \tag{2}$$

where V_{ac} is the output AC voltage.

The efficiency η of a rectifier is the ratio of the output (DC) power to the input (AC) power:

$$\eta = \frac{P_{\text{dc}}}{P_{\text{in}}} = \frac{V_{\text{dc}}^2/R_L}{V_{\text{in}}^2/(R_d + R_L)} = \frac{V_{\text{dc}}^2}{R_L V_{\text{in}} I_{\text{in}}} \tag{3}$$

where V_{in} is the RMS value of the input AC voltage, I_{in} is the RMS value of the input AC current and R_L is the load resistance.

The output voltage from the full-wave bridge rectifier across the load resistance can be stabilised by adding a capacitor in parallel to the load resistance. This is known as a filter and stabilises the output DC voltage across the load resistance.

3 Materials Required

(i) Breadboard, (ii) diodes (4 identical), (iii) capacitors (3 of different capacitance), (iv) resistors (4 of different resistance), (v) multimeter, (vi) connecting wires, (vii) oscilloscope, (viii) step-down transformer/function generator.

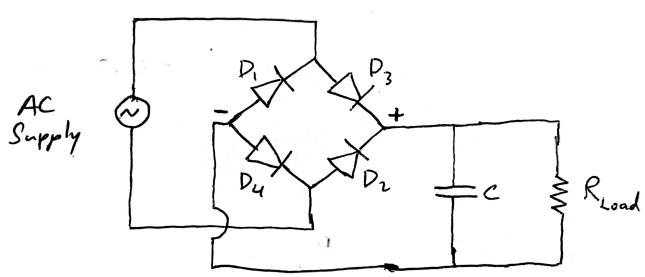


Figure 1: Circuit diagram of a full-wave bridge rectifier. The AC voltage is provided by the AC supply whereas the rectified DC voltage is obtained across R_{Load}. The capacitor C acts as a filter which is optional but stabilises the output DC voltage.

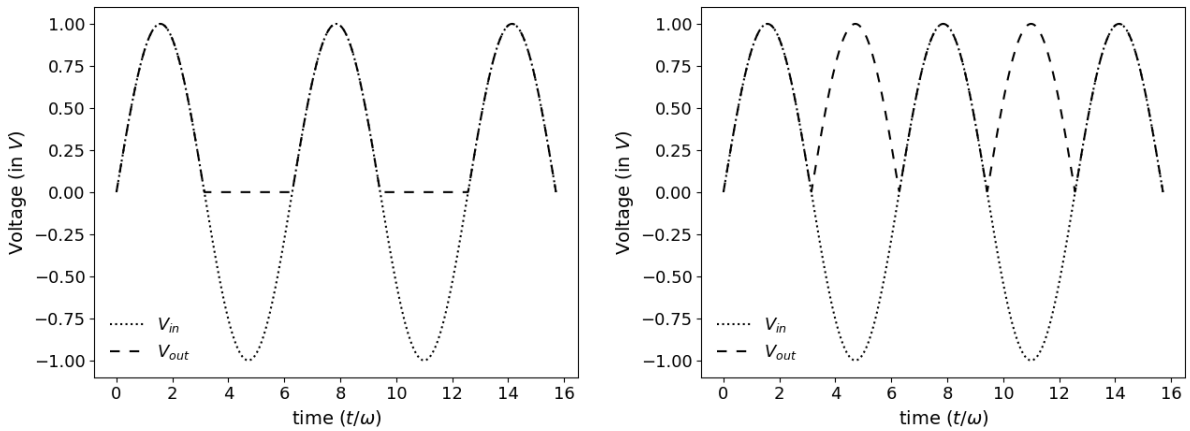


Figure 2: Rectified signals from a half-wave rectifier (left) and a full-wave rectifier without filter (right) as predicted by theory. Input AC voltage is given using dotted plot whereas the output signal is given as dashed plot.

4 Observations

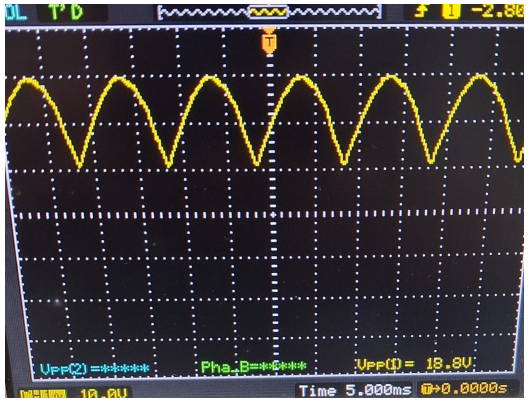
Code number of diode = IN4007
Input AC voltage: $V_{\text{in}}(\text{rms}) = 8.00\text{V}$

Table 1: Full-wave rectifier observations without filter

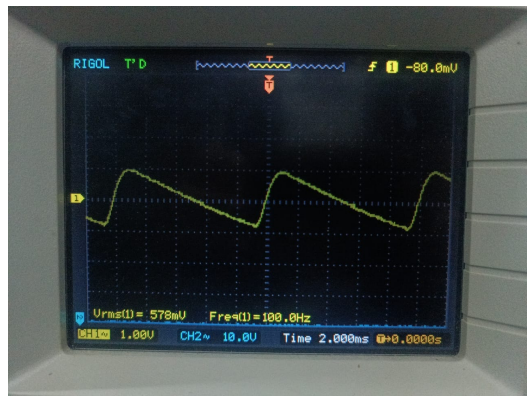
Sl. No.	Load R_L (kΩ)	Input Current I_{ac} (mA)	Output Voltage V_{ac} (V)	V_{dc} (V)	$2V_{max}/\pi$ (V)	Ripple Factor r	Efficiency η $V_{dc}^2/(R_L V_{in} I_{in})$ (%)
1	0.549	12.00	3.17	5.74	7.20	0.55	62.5
2	9.88	0.69	3.22	6.03	7.20	0.53	66.6
3	1.008	6.60	3.18	5.80	7.20	0.55	63.2
4	4.68	1.44	3.20	5.98	7.20	0.54	66.3

Table 2: Full-wave rectifier observations with filter

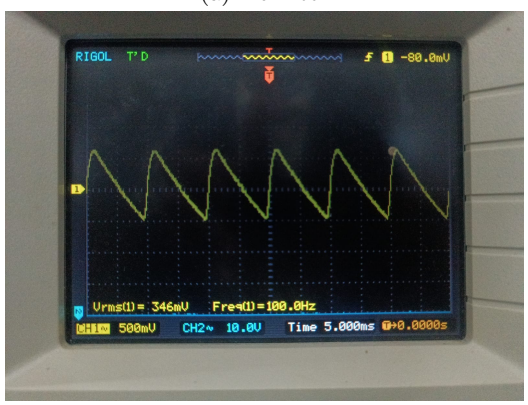
Filter	Sl. No.	Load R_L (kΩ)	I_{ac} (Input) (mA)	Output Voltage V_{ac} (V)	V_{dc} (V)	Ripple Factor r
$C = 16.72\mu$F	1	1.008	15.5	1.076	8.06	0.13
	2	0.549	22.5	1.562	7.36	0.21
	3	9.88	2.8	0.159	9.70	0.02
	4	4.68	5.6	0.32	9.48	0.03
$C = 39.5\mu$F	1	1.008	19.4	0.594	8.72	0.07
	2	0.549	29.3	0.942	8.12	0.12
	3	9.88	2.8	0.076	9.78	0.01
	4	4.68	5.8	0.152	9.54	0.02
$C = 9.68\mu$F	1	1.008	13.0	1.506	7.60	0.20
	2	0.549	18.7	2.051	6.88	0.30
	3	9.88	2.7	0.262	9.62	0.03
	4	4.68	5.0	0.506	9.12	0.06



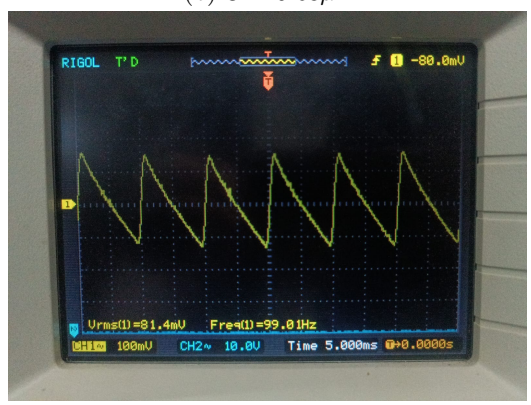
(a) No filter



(b) $C = 9.68\mu$F



(c) $C = 16.72\mu$F



(d) $C = 39.5\mu$F

Figure 3: The waveform of the rectified signal viewed in the oscilloscope for different filters and without any filter.

5 Results

The ripple factor r and the efficiency η for the rectifier in tables 1 and 2 is calculated using equations 2 and 3 respectively.

From table 1 for without filter case, the ripple factor r and the efficiency η increases with an increase in the load resistance R_L.

From table 2 for with filter case, if the resistance is kept constant, the ripple factor r is smaller for large capacitance as compared to small capacitance. Also, if the capacitance is kept constant for the filter, the ripple factor r decreases with the increase in the load resistance R_L.

From the waveforms in figure 3, it can be verified that the waveform of the output signal shows the characterisitics as predicted by theory.

6 Discussion

The experiment confirmed that the given circuit for the full-wave bridge rectifier can rectify an input AC voltage into a DC voltage. An increase in the capacitance of the filter decreases the ripple factor of the rectified DC voltage as predicted. The efficiency of the rectifier is around 60-70% which is less than the theory predicted $\eta \approx 81.1\%$ efficiency. This may be because the transformer has some energy loss during its functioning or in the resistance of the wires and breadboard. Increasing the load resistance increases the voltage drop across the load which may explain the increase in efficiency η with an increase in the resistance (see equation 3).

7 Conclusion

The full-wave bridge rectifier is thus constructed and an analysis of its output is given in tables 1 and 2. The ripple factor r decreases with the increase in the capacitance of the capacitor in shunt with the load resistance.

8 Precautions

1. Ensure that the diodes conduct in only one direction and not in reverse bias.
2. The capacitor (if polar) must be connected with the correct polarity in the circuit.
3. Both the input and output voltage should not be measured at the same time as to avoid two seperated grounds.
4. The wires of the step-down transformer should not be shorted to avoid damage to the transformer.